

1(a)(i)	<i>independent</i> concentration / molarity, of sucrose (solution) ; <i>dependent</i> change in mass / AW ;	2
1(a)(ii)	1 five stated concentrations and units (from 1.0 mol dm ⁻³ downwards ; 2 <i>idea of</i> evenly spaced ; 3 method for dilution shown for at least 2 intermediates ;	3

Question	Answer	Marks
1(b)	any six from: 1 <i>idea of</i> tissue cut to (approximately) same / equal / AW, size / mass ; 2 tissue taken from the same, pepper / age of pepper / state of pepper ; 3 set up containers / named containers, containing different sucrose concentrations ; 4 same / stated, volume of sucrose solution or enough (sucrose solution) to, cover / immerse, the tissue ; 5 covering container (to prevent evaporation) ; 6 leaving tissue (in solution) for same time ; 7 method of maintaining a constant temperature ; 8 mass of tissue measured / recorded, before and after immersion ; 9 tissue dried (with paper towel / AW) before weighing ; 10 minimum of 3 repeats and calculate mean ; 11 low risk or medium risk qualified ;	6
1(c)(i)	$\frac{\text{final mass} - \text{initial mass}}{\text{initial mass}} \times 100$ 1 of: (initial) masses, are / maybe, variable / not the same ; allows qualified comparisons to be made (between masses) ;	2

Question	Answer	Marks
1(c)(ii)	1 x-axis labelled and unit ; 2 line crosses x-axis (only once) and trend downwards (from left to right) ; 3 <i>idea of</i> indication that interception on the x-axis shows water potential ;	3
1(d)(i)	–860kPa ;	1
1(d)(ii)	<i>any three from:</i> <i>all idea that</i> 1 difficult to, place / release, drop at same point in solution ; 2 judgement (of position of drop) by eye / method 2, is, subjective / qualitative / subject to human error ; 3 measurement of mass / method 1, is, objective / quantitative ; 4 method 1 can (more easily) plot a graph (to estimate water potential) ; 5 drop might disperse / disintegrate / is unstable /AW ; 6 method 2 does not take into account different masses of tissue (before initial soaking) ; 7 importance of mixing solution once tissue removed in method 2 ; 8 method 2 has not been repeated ;	3